

Aman Kumar

Backend Developer

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Summary

Backend Developer with 3+ years of experience designing scalable backend systems, REST APIs, and data-intensive applications. Skilled in Python, Django, PostgreSQL, system design, and cloud architecture (AWS). Experienced in collaborating with diverse technical teams to engineer high-performance cloud-native applications handling millions of records and enterprise workloads.

Professional Experience

Genpact

Aug 2023 – Present

Software Developer

Bengaluru, India

- Partnered with cross-functional squads to architect and scale high-throughput microservices in Python and Django, successfully processing **over 500,000 requests monthly** with low latency.
- Refined complex SQL queries, indexing strategies, and database access patterns for large-scale datasets, reducing query execution time by **60%**.
- Spearheaded the design of a scalable caching layer using Redis and the Cache-Aside pattern, implementing a least recently used eviction algorithm to reduce database read load by **40%** on high-traffic endpoints.
- Client Project: GE Vernova (Data Access Portal)** | *Python, Django REST API, SQL, Cloud Datasets*
 - Developed a financial visualization platform with a natural language interface, empowering **over 100 users** to generate custom JSON templates and **accelerating analysis by 30%**.
 - Constructed SQL to JSON query translation and validation pipelines using cloud datasets, improving query accuracy and reducing manual validation effort.
 - Partnered with client stakeholders to migrate reports from EY to GE Vernova by programming automated scripts, successfully importing **over 130 blank templates** and **10,000 associated reports**, streamlining the transition with minimal downtime.
- Client Project: GE Vernova (Data Archival Portal)** | *Python, Django REST API, AWS S3, Athena, OpenSearch*
 - Architected a low-cost archival platform to migrate metadata to AWS S3 cold storage, cutting **storage costs by 80%** across 230 enterprise projects.
 - Implemented row-level security for structured tables queried via **AWS Athena**, enabling secure filtering and automated data exports across **6 million files**.
 - Integrated AWS OpenSearch for full-text search across **3 million encrypted files**, building paginated APIs to ensure stable and high-performance data retrieval.
- Client Project: GE Vernova (Vernova Data Portal)** | *Python, Django REST API, SQL, Tableau, AWS*
 - Led the backend revamp of the legacy portal into the Vernova Data Portal, coordinating with frontend developers to establish a hierarchical domain-based user access model and embed a custom chatbot alongside Tableau and Spotfire reports using iframes.

Enquero Global LLP (A Genpact Company) [Certificate]

Mar 2023 – Jul 2023

Software Engineering Intern

Bengaluru, India

- Enhanced a fitness platform by engineering RESTful APIs for subscription features, session booking, and vitals monitoring using **Python and Django**, increasing user engagement by **25%**.

Skills

Languages: Python, SQL, C++

Backend Frameworks: Django, Django REST Framework, FastAPI

Databases & Caching: PostgreSQL, MySQL, Redis, AWS Athena, SQL Optimization

Cloud & DevOps: AWS, Docker, OpenSearch, Linux

Background Processing: Asynchronous Workflows, Message Queues, Task Scheduling

Core CS & Architecture: System Design, Distributed Systems, REST APIs, Data Structures & Algorithms

Certificates and Achievements

Microsoft Certified: Azure Fundamentals (AZ-900) [Certificate]

Agentic AI - Fundamentals [Certificate]

Generative AI - Fundamentals [Certificate]

Databricks Certified Data Engineer Associate [Certificate]

HackWithInfy 2022 Qualified

Coding Profiles: LeetCode — GeeksForGeeks

Education

Nitte Meenakshi Institute of Technology

2019 – 2023

Bachelor of Engineering in Computer Science — CGPA: 8.51/10.0

Bengaluru, India